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1. PART 1: INTRODUCTION AND BACKGROUND OF THE SCANDAL

1.1. Introduction

1.1.1. Introduction of the company

Once a part of Fiat Chrysler Automobiles, it sold to Stellantis in 2021 (Psa Group) and by 2015, it accounted for more than 4.6 million cars sold worldwide (Statista, 2021). The Jeep Cherokee, one of the company's midsize SUVs, used FCA's Uconnect system, tying navigation, media, and diagnostics functions to a Sprint mobile data plan. Through this setup, each car received its own unique IP address, which allowed control systems to be accessed externally (Greenberg, 2015a). While cars were designed as digital devices built on code and web links, more traditional design methods did not protect against the new risks posed by constant connectivity.

1.1.2. Introduction to Cyber Threats and Risks

The various types of Cyber threats and risks are as mentioned.

Threat Type	Description
Remote Code Execution (RCE)	An attacker finds vulnerabilities in software to gain remote access to alter commands and act maliciously on that system (Bugcrowd, 2021).
CAN Bus Spoofing	For an unauthenticated system, signals are forged and sent to that network to have its systems execute false commands (ENISA, 2021).
Man-in-the-Middle (MitM)	An attacker would inject themselves and modify the intercepted signals in the process (ENISA, 2021).
Denial of Service (DoS)	Critical components are flooded with bot traffic until they become unresponsive to legitimate inputs .
Supply Chain Attacks	Malicious or vulnerable code is embedded by a third-party supplier and propagates silently through the target system.

Table 1: Types of Cyber Threats.

1.2. Background of the scandal

In July 2015, cybersecurity researchers Charlie Miller and Chris Valasek remotely took control of a 2014 Jeep Cherokee travelling on a St. Louis highway, with Wired journalist Andy Greenberg in the driver's seat (Greenberg, 2015a). From a laptop miles away, they manipulated the climate controls, audio, windshield wipers, and transmission, then cut the engine at highway speed and interfered with braking at low speed (Greenberg, 2015b).

The attack exploits several of the vehicle's architecture flaws: Miller and Valasek obtained access to Uconnect's D-Bus port 6667, which is open to a Harman-built QNX operating system. TCP traffic was filtered to accept data that didn't match the schema, then they used it to directly communicate with the vehicle's CAN bus (Valasek, 2015).

Not a single vulnerability, but an ineffective application of secure-by-design principles throughout FCA's design. There was no network authentication, no system isolation, and no penetration testing when Checkoway et al. had shown exactly this kind of attack in published research years earlier. Miller and Valasek also privately disclosed the flaws to FCA nine months before going public. There was no fix, making disclosure mandatory (Greenberg, 2015a).

Under pressure from the Obama administration, FCA recalls 1.4 million vehicles. Uconnect is not up-to-date over the air, and patches have to be manually installed by owners via USB. Senators Markey and Blumenthal introduced the SPY Car Act shortly thereafter. The pattern appears to be continuing: As Upstream Security reported in 2023, cybersecurity incidents in the automotive industry increased by 225% over four years, with most of the cases occurring via remote attacks in 2022 (Skeath, 2015).

Question	Details
Who?	Charlie Miller and Chris Valasek both have a background in auto vulnerability research (Valasek, 2015). Miller previously was an NSA analyst with a Ph.D. in Mathematics, and Valasek also specialized in vehicle security at IOActive. Charlie Miller and Chris Valasek, the Director of Vehicle Security Research at IOActive, worked together. Andy Greenberg, a journalist for Wired, agreed to the opportunity of being the driver during the test.
What?	Miller and Valasek controlled a 2014 Jeep Cherokee remotely using its Uconnect system while it was driving on a live highway. They changed climate and audio settings, activated the wipers, killed the engine, disabled brakes, and even drove the vehicle without ever stepping inside it (Greenberg, 2015a).
When?	Miller and Valasek told FCA about the vulnerability in October 2014. FCA did nothing. Nine months later, on 21 July 2015, the demonstration went public through Wired and was presented at DEF CON 23 (Valasek, 2015).
Where?	The Jeep was on Interstate 64 in St. Louis, Missouri. Miller and Valasek were sitting in a house nearby, nowhere near the car (Greenberg, 2015a).
Why?	FCA had brushed off the private report. Earlier academic research, including work by Checkoway et al., had also been ignored by the industry. Going public was the only option left that would actually force a response (Greenberg, 2015b).
How?	They came in through Sprint's cellular network and hit port 6667 on Uconnect's QNX system. That port took malformed TCP requests without asking for any login. From there, nothing separated the infotainment unit from the CAN bus, so their messages went straight to the ECUs controlling the brakes, engine, and steering (Valasek, 2015).

Table 2: Table in Wh question format.

1.2.1. Past similar test cases

The 2015 exploit did not emerge in isolation. A clear pattern of escalating automotive cyber-risk preceded it. The evolution of the failure with each test case is given in [Appendix A](#).

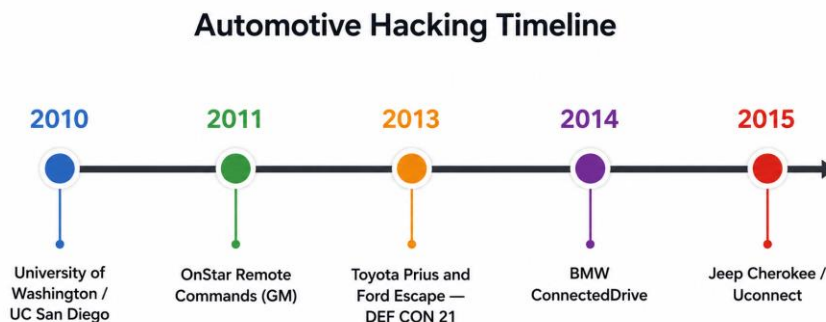


Figure 1: Automotive Hacking Timeline.

1.2.2. Key Statistics from the case

Jeep Hack (2015) – Key Statistics Overview

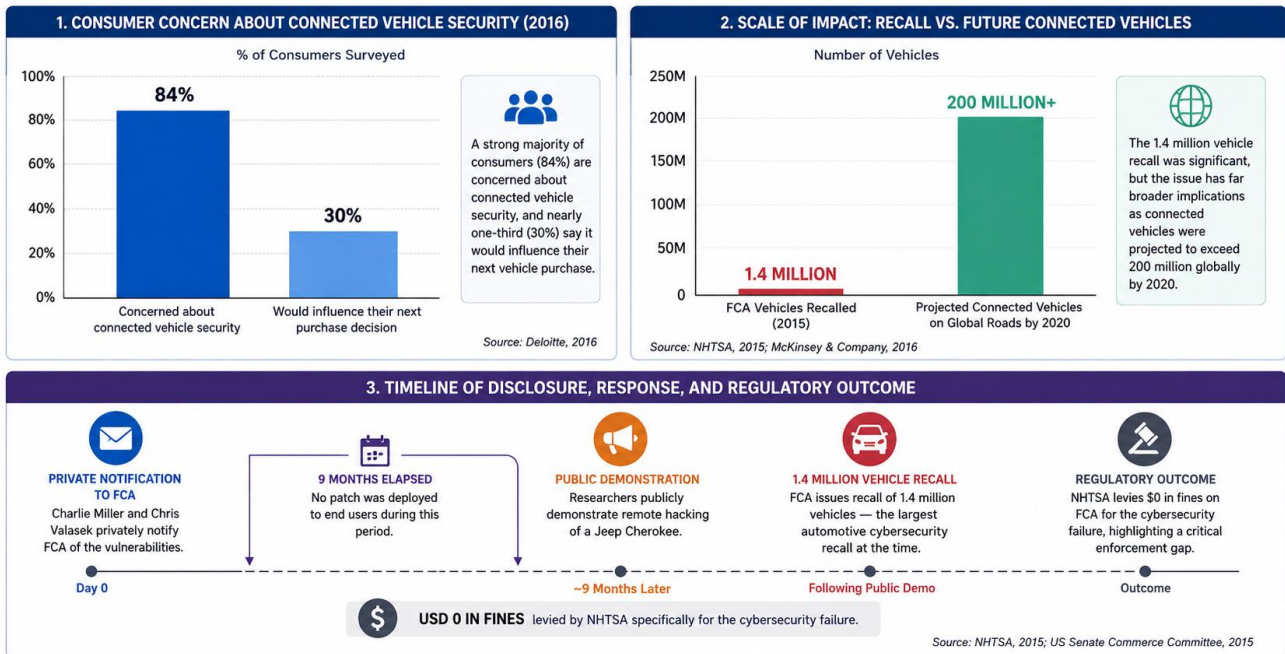


Figure 2: Key Statistics Overview.

Sources: (NHTSA, 2016), (Greenberg, 2015b), (U.S. Senate Committee, 2015)

For brief explanation, please refer to [Appendix B](#).

2. PART 2: SOCIAL ISSUES

2.1. Background on Social Issues

Social issues arising from this case extend well beyond consumer privacy. They touch the structure of public safety, democratic access to technology, and the psychological contract between the individual and the machines they depend on daily. The following five issues are identified as distinct and significant social consequences of the Uconnect vulnerability.

2.2. Identified Social Issues

Below mentioned are some of the identified social issues that commence to exist after implications mentioned above are seen:

2.2.1. Social Issue 1: Weaponization of Mobility Dependency

Our lives are essentially lived on cars. An unpatched car is not just a digital data breach but also a dangerous vehicle. While hacking can be done without any immediate effect, it is likely to be traveling close to people and objects. A flaw in the Uconnect system exposed a way to turn a simple device into a deadly weapon. In countries like Nepal where the WHO estimates road death tolls at 19.9 per 100,000 people, the possibility of this kind of hijacking could make already difficult situations even more dangerous (WHO, 2018). Connectivity is sold as progress, but when it is not the most vulnerable people.

2.2.2. Social Issue 2: Digital Divide in Automotive Cyber-Risk

The majority of the 1.4 million recalled vehicles were mid-range vehicles bought by working and middle-class families with loans and financing. These owners had no way to know if their vehicle was patched, and the recall of a USB drive mailed to owners presumed that these people have digital literacy and were able to get the package in time (Greenberg, 2015b). Older vehicles were spared, unconnected. This creates a paradox, where the consumers who purchased the most technologically advanced vehicles were at risk, but were the least prepared to respond. Zuboff reported that digital technology is helping those who have the best benefits of digital technology while its negative effects are felt across the disadvantaged (Zuboff, 2019).

2.2.3. Social Issue 3: Erosion of Public Trust in Connected Infrastructure

One failure shook faith beyond a company. V2X (Vehicle-to-Everything) links exist alongside self-driving cars and urban signal networks, and trust forms where machines are expected to perform. According to KPMG in 2016, a lack of safety outweighs concerns about engineering capacity (KPMG, 2016). The incident with the Jeep raised the stakes, and delayed the allocation of transport tech funding in the U.S. and Europe.

2.2.4. Social Issue 4: Media Ethics and Non-Consenting Bystanders

The live demonstration was on a public interstate highway. Miller and Valasek and others on the road were unaware that a vehicle in their own paths was being disabled for a media stunt. When Miller and Valasek cut the Jeep's transmission at highway speeds, the vehicles behind it suddenly felt a sudden but unprovoked deceleration from a car they believed was being driven by a human (Greenberg, 2015a). This raises a social ethics question that has received limited attention: how does a security researcher's right to show a vulnerability affect those who are not consenting to such demonstrations? Wired's decision to publish the demonstration live further exacerbated the issue (Greenberg, 2015a).

2.2.5. Social Issue 5: Normalisation of Surveillance Culture in Transportation

FCA and other manufacturers sped up the deployment of over-the-air (OTA) update systems as a security response. Although this was necessary, OTA systems also increased the amount of information that every vehicle had to transmit by radio to manufacturer servers: position, speed, driving behaviour, and diagnostics. As the Uconnect hack forced the adoption of OTA systems, it also led to a new paradigm of surveillance of vehicles. Consumers paid the price of security for a permanent loss in their privacy of mobility, a cost that received little public debate (Zuboff, 2019).

3. PART 3: ETHICAL ISSUES

3.1. Background on Ethical issues

Ethical analysis requires application of multiple normative frameworks to evaluate actions and omissions of FCA, its development team, and the researchers. The case sits at the intersection of utilitarian calculation of net societal benefit, Kantian deontological duties of care and disclosure, virtue ethics concerns regarding professional integrity, and contractarian obligations arising from the implicit social contract between technology vendors and consumers (Michael J, 2020).

3.2. Identified Ethical issues

3.2.1. Ethical Issue 1: Informed Consent Violation in Live Demonstration

Andy Greenberg was the reporter on the demonstration that the researchers expected to have a hacking demo. He was not informed that his brakes would be disabled at highway speed or that his transmission would be cut in live traffic (Greenberg, 2015a). This is a violation of informed consent as defined in the research ethics literature: consent must be specific and not general (Beauchamp, 2013). Greenberg consented to a technology demonstration, not a near-accident at 70 mph. The researchers used a consenter as an experiment without fully telling the parameters of the experiment.

3.2.2. Ethical Issue 2: Coordinated Disclosure Ethics: Escalation or Exploitation?

Miller and Valasek followed the usual protocol for responsible disclosure, notifying FCA nine months before publication. FCA replied dismissively, as stated by the researchers (Valasek, 2015). Is it ethical to push such a high-profile release on an active website and then publish a full technical white paper with exploit code, or weaponizing such a news release to do more harm to FCA? The dual role of the release: the most potent lever of industry reform and the ultimate attack guide for malicious actors remains unresolved ethical tension in the security community.

3.2.3. Ethical Issue 3: Manufacturer's Prior Knowledge and Duty of Care

Senator Edward Markey's report on security in 2014 found that all sixteen out of the 16 automakers, including Chrysler, did not have basic intrusion detection and that Chrysler's connected vehicle systems did not detect odd signals being transmitted over the CAN bus (Markey, 2015). FCA was advised of the vulnerability and chose not to act. FCA has left a knowingly defective product on the market long before the conditions that necessitated the demonstration of 2015.

3.2.4. Ethical Issue 4: Design Ethics: CAN Bus Architecture and Deliberate Segmentation Failure

Bosch's design of the CAN bus protocol, created in 1986, does not include an authentication mechanism because it was created for a closed, pre-connectivity world (GmbH, 1991). Uconnect engineers connected the infotainment head unit to the same CAN bus as the safety-critical ECUs without any firewall, intrusion detection (ID) code or messages authentication code. The bad decision to connect a safety-critical system to an unauthenticated bus is engineering negligence that can cause foreseeable harm.

3.2.5. Ethical Issue 5: Corporate Priority Inversion: Features Over Safety

Uconnect's user-facing features, such as an 8.4-inch screen, SiriusXM, and 4G Wi-Fi hotspot, were crucial to FCA's marketing differentiation strategy. The security budget grew in opposite proportion to the feature budget, highlighting the "security by obscurity via complexity" argument of Nissenbaum, where an intuitive interface is considered too complicated for hackers to exploit (Nissenbaum, 2004). The ethical failure is organisational; when a manufacturer treats cybersecurity as a cost and infotainment as a profit centre within a safety product, spending resources can affect life-or-death decision.

3.3. Philosophical comparison table

The following table provides a comparison of ethical issues with 4 major philosophies.

Ethical Issues	Deontology	Utilitarianism	Virtue Ethics	Rights-Based
Live Highway Demo	Bystanders never consented to the risk.	Thin net benefit, one journalist's risk vs. millions safer long-term.	A prudent researcher tests in controlled settings first.	Greenberg and bystanders both lacked proper disclosure.
Full Technical Disclosure	Duty to warn outweighs commercial interests.	Publishing accelerates patching but only if the window is short.	Notify manufacturer first, public second.	Public safety rights trump technical transparency.
FCA Ignoring Prior Warnings	Knowing in 2014 and staying silent violated basic duty.	1.4M exploitable vehicles inaction had a daily cost.	Silence to protect reputation is cowardice, not caution.	Buyers had a right to a safe product from day one.
Dual-Use Research	Exploit code shouldn't publish before a universal patch exists.	Net utility unclear depended entirely on who read it first.	Researchers underweighted copycat harm.	Safety rights precede the security community's transparency interests.
Sprint Complicity	Sprint co-owned the attack chain; monitoring was its obligation.	Blocking cost little; inaction cost users far more.	Not watching your own network isn't neutral, it's negligent.	Users trusted Sprint not to route attacks to their cars.

Table 3: Philosophical Comparison Table.

4. PART 4: LEGAL ISSUES

4.1. Background on Legal Issues

The legal landscape implicated by the incident spans criminal liability, product liability, regulatory enforcement, third-party developer liability, and European law. In the scenario where exploitation was malicious rather than demonstrative, consequences would have fallen across multiple parties simultaneously.

4.2. Identified Legal Issues

4.2.1. Legal Issue 1: Criminal Liability [18 U.S.C. § 1030(a)(5) - CFAA]

- Criminalises unauthorised access and intentional transmission of damaging code.
- Violated: Knowingly exploiting Uconnect's cellular port to transmit control commands (Valasek, 2015).
- Penalty: Up to 10 years imprisonment; up to 20 years where death results (US Department of Justice, 2015).

4.2.2. Legal Issue 2: Product Liability [Restatement (Third) of Torts § 2 - CPSA]

- Products "unreasonably dangerous" due to design defects attract strict liability (American Law Institute, 1998).
- Violated: Uconnect's unprotected CAN bus connection constitutes a design defect.
- Penalty: FCA issued a 1.4 million vehicle recall at an estimated cost of USD 105 million (Skeath, 2015).

4.2.3. Legal Issue 3: NHTSA Regulatory Violations [49 U.S.C. § 30165]

- NHTSA's 2016 Cybersecurity Best Practices confirmed cybersecurity as a motor vehicle safety issue (NHTSA, 2016).
- Violated: FCA's failure to remediate a known vulnerability breached NHTSA safety standards (US Department of Transportation., 1998).

- Penalty: FCA paid USD 105 million — the largest civil penalty in NHTSA history at the time.

4.2.4. Legal Issue 4: Supply Chain Negligence [ISO/SAE 21434:2021]

- Negligence per se applies where established industry standards are ignored.
- Violated: Uconnect developers' failure to follow ISO/SAE 21434 exposes them to joint and several tort liability alongside FCA (ISO, 2021).
- Penalty: No penalty was imposed; however, joint and several liability could extend recall costs to third-party developers (ISO, 2021).

4.2.5. Legal Issue 5: European Legal Exposure [GDPR Article 32 / UNECE WP.29 R.155]

- Uconnect collected and transmitted data without adequate security measures.
- Violated: GDPR Article 32 - failure to implement appropriate technical security measures (European Parliament, 2016).
- Violated: UNECE WP.29 R.155 - absence of a certified Cybersecurity Management System (UNECE, 2021).
- Penalty: GDPR fines up to 4% of global annual turnover; WP.29 R.155 non-compliance bars vehicle type approval in EU, UK, Japan, and South Korea from July 2022.

5. PART 5: PROFESSIONAL ISSUES

5.1. Background on Professional Issue

Professional issues arise when practitioners act in violation of their codes through active misconduct, negligence, or organisational complicity. The following five issues are analysed in the hypothetical scenario where the hackers chose malicious exploitation rather than responsible disclosure.

5.2. Identified Professional Issues

5.2.1. Professional Issue 1: Obligation to Avoid Harm - FCA's Architectural Decision [ACM Code 1.2]

- **ACM Code 1.2 states:** "Computing professionals should avoid creating systems or products that pose unnecessary risk to the public." (ACM, 2018)
- **Comparison:** ACM Code 1.2 requires professionals to avoid systems that create unnecessary public risk. FCA engineers violated this principle by connecting infotainment systems directly to safety-critical CAN networks without proper isolation or authentication mechanisms (ACM, 2018).

5.2.2. Professional Issue 2: Paramount Duty to Public Safety [IEEE Code of Ethics - Article I, Clause 1]

- **IEEE Code Article I.1 requires:** Engineers shall "hold paramount the safety, health, and welfare of the public" in the performance of their professional duties (IEEE, 2020).
- **Comparison:** IEEE Article I.1 requires engineers to prioritise public safety, health, and welfare above commercial interests (IEEE, 2020). FCA failed to uphold this duty by approving deployment of a vulnerable vehicle architecture that exposed drivers to foreseeable physical harm.

5.2.3. Comprehensive Risk Evaluation Requirement [ACM Code 2.5]

- **ACM Code 2.5 states:** "Computing professionals should give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks." (ACM, 2018).
- **Comparison:** ACM Code 2.5 requires thorough evaluation of systems and associated risks before deployment. FCA failed to conduct sufficient cybersecurity assessment and penetration testing on Uconnect's integration with the CAN bus despite existing research demonstrating similar attack possibilities (Greenberg, 2015a).

5.2.4. Public Interest Obligation [BCS Code of Conduct - Section 1(a)]

- **BCS Code Section 1(a) states:** "Have due regard for public health, privacy, security and wellbeing of others and the environment." (BCS, 2022).
- **Comparison:** BCS Section 1(a) requires professionals to protect public health, security, privacy, and wellbeing (BCS, 2022). Uconnect developers failed to adequately address known security weaknesses, exposing consumers and the wider public to unnecessary safety risks.

5.2.5. Responsibility Not to Certify Sub-Standard Systems [SE Code of Ethics - Principle 3.12 (ACM/IEEE)]

- **SE Code Principle 3.12 states:** Software engineers shall "not influence others to undertake any product or project they believe will fail, be sub-standard, involve a risk, or might create a risk to others." (ACM, 2018), (IEEE, 2020).
- **Comparison:** SE Code Principle 3.12 states that software engineers should not approve systems they believe may create risks to others. FCA's release of Uconnect despite exploitable vulnerabilities in a safety-critical environment reflected a failure to prevent deployment of an unsafe system (ACM, 2018) (IEEE, 2020).

6. PART 6: CONCLUSION AND PERSONAL REFLECTION

6.1. Conclusion

The Jeep Cherokee exploit of 2015 is no anomaly. It is a metaphor for products which mix engineering safety with digital software security. Not so. Every control that a driver's foot puts on the brake pedal, or a stranger on Sprint's cellular network, has sent to the ECU passes through the same CAN bus. Lose trust, mobility risk, normalised surveillance, these will outlive the patch. Failures in ethics, lack of consent in testing, hidden warnings, cheap design reveal how the automotive industry treats itself. The legal landscape in 2015 was apathetic: CFAA was for computers, not for cars, and NTMVSA was made for mechanical faults, not software weaknesses. This case resulted in the largest global automotive cybersecurity rule in history (UNECE WP.29 R.155). That is a victory for the industry.

6.2. Personal Reflection

If I was Miller and Valasek, I would have followed coordinated disclosure. But, I would not have released the exploit code until after I had patched all affected vehicles. My ethics decision would then be guided by the Four-Component Model of ethical decision making: (1) Recognise the moral issue: the CAN bus vulnerability poses a risk to lives; (2) Make a moral judgement: the public has a right to safety that outweighs FCA's corporate reputation; (3) Establish moral intent: the goal should be industry reform, not corporate harm; and (4) Act morally: release information about the vulnerability, with a cure period of 90 days, patches are installed before technical details are revealed. According to ACM Code 2.5 and IEEE Article I.1, researchers have a responsibility to the public, and not for the comfort of the manufacturer. But, under ACM Code 1.2, researchers have a moral obligation to not create new risks by publishing the complete attack methodology before security measures are in place. The ethical path is more than secrecy or maximum exposure: it requires precision, timing, and real priority for the public good of the research over prestige of the researchers.

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8. Appendix

8.1. Appendix A: Brief Explanation of Past Similar Test Cases.

Year	Case	What Happened
2010	University of Washington / UC San Diego	Koscher et al. were the first to prove remote car hacking was real. They got in through Bluetooth and a malformed audio file played through the CD player, then manipulated braking and engine functions. The industry largely ignored it (Karl Koscher, 2010).
2011	OnStar Remote Commands (GM)	Researchers showed that OnStar's remote command system could be intercepted and replayed. An attacker could unlock doors or locate a vehicle without the owner's knowledge. GM patched it quietly.
2013	Toyota Prius and Ford Escape, DEF CON 21	Miller and Valasek, physically plugged into the OBD-II port, demonstrated brake disabling, sudden acceleration, and steering control on both cars. Physical access was still required, but it proved the CAN bus was completely undefended once you were in (Valasek, 2013).
2014	BMW ConnectedDrive	Germany's ADAC automobile club found that BMW's ConnectedDrive portal could be spoofed using a fake mobile network. Attackers could unlock any of the 2.2 million affected vehicles remotely with no authentication required (ADAC, 2015).
2015	Jeep Cherokee / Uconnect	Miller and Valasek went fully remote, no physical access, no proximity required, from an ordinary house over Sprint's cellular network. Every previous case had required either a cable, Bluetooth range, or a compromised update server. This one needed none of that (Valasek, 2015).

[Go back.](#)

8.2. Appendix B: Key Statistics

- 1.4 million FCA vehicles recalled, the largest automotive cybersecurity recall at the time (NHTSA, 2016).
- 9 months elapsed between FCA's private notification and the public demonstration, during which no patch was deployed to end users.
- 84% of consumers surveyed in 2016 expressed concern about connected vehicle security, with 30% stating it would influence their next purchase decision.
- 200+ million connected vehicles were projected to be on global roads by 2020, making the systemic implications of the Jeep hack far beyond a single product recall.
- USD 0 in fines were levied on FCA by NHTSA specifically for the cybersecurity failure, revealing a critical enforcement gap in existing regulation (U.S. Senate Committee, 2015).

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